ESTIMATING THE CAUSAL EFFECT OF AI-TOOL/CHATGPT USE (X) ON HOURS SPENT ON SCHOOLWORK (Y) USING PEARL'S BACK-DOOR CRITERION

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Abstract. Generative artificial intelligence (AI) tools (e.g. ChatGPT, Claude) have been rapidly adopted in education since the public release of ChatGPT in late 2022. A synthesis of eight recent studies (2024 - 2025) shows that ChatGPT use causally reduces the time required for specific academic tasks through cognitive load reduction and information access efficiency. However, the net effect on total hours spent on schoolwork is contingent on several confounding factors: workload (W), a pre-treatment note collecting prior ability, motivation and course difficulty (A), AI literacy (L) and two post-treatment mediators: (i) M_1–task efficiency or time saved and; (ii) M_2 AI-induced procrastination or overreliance. In this paper, we attempt to answer the question: **What is the causal effect of using ChatGPT (X) on the number of hours students spend on independent study and schoolwork (Y)?** A primary survey (n = 15) conducted associated high AI use with studying 7.1 h/week more. A secondary analysis of the Colombian study-habits survey (n = 357), where the back-door-adjusted estimate was also positive (β = 0.216 log-odds, SE = 0.132, p = .10; equivalently +0.16 h/week, SE = 0.094, p = .09). Therefore, the method proposed estimates the causal effect of AI-tool/ChatGPT use (X) on hours spent on schoolwork (Y) using Pearl's back-door criterion, while showing that a simple bivariate t-test can't answer the causal question.

1. Introduction

Generative AI tools have spread through education quickly since ChatGPT was released publicly in late 2022, and the obvious question for a student is whether using them changes how long schoolwork takes. Two mechanisms are proposed in the literature and they point in opposite directions.

Under *substitution*, AI absorbs work the student would otherwise have done alone, so heavier use goes with fewer independent study hours. Under *complementarity*, AI raises the productivity of study time and is adopted preferentially by students already investing heavily, so heavier use goes with more hours. Existing work supports the first mechanism at the level of a single task without settling the aggregate. Dahri *et al.* [4] report that ChatGPT use predicts perceived reductions in time spent on traditional learning resources, but their outcome is a perception item about resource substitution rather than total schoolwork hours. Li *et al.* [6] log GenAI-assisted problem solving in fine detail for 279 students, and their timestamps measure time on a two-week assignment rather than a student's weekly total. Contractor and Reyes [3] survey 634 students and collect weekly study hours alongside GenAI frequency, which is the closest variable match available, but respondent-level microdata is not distributed with the paper. Task-level time savings are therefore better evidenced than any net effect on hours, and the net effect is what a student actually experiences.

Proper AI usage is important because it helps people work more efficiently, generate ideas, solve problems and improve the quality of their work. However, AI should be used responsibly rather than relied on completely. Users must check information for accuracy, protect personal data, respect copyright and avoid using AI to cheat or mislead others. Good prompts and careful judgement are also needed to produce useful results. AI is most effective when it supports human creativity and decision-making, not when it replaces independent thinking. By using AI ethically and thoughtfully, we can enjoy its benefits while reducing risks and creating trustworthy outcomes.
[1]

This study aims to determine whether using generative AI tools for schoolwork changes the number of hours a student spends studying independently. The research will: (i) Measure weekly AI-tool use and independent study time among Ngee Ann Polytechnic students through a primary survey (n = 15); (ii) Test whether mean independent study time differs significantly between low and high AI-use students, using a two-sample t-test at $\alpha = 0.05$ and; (iii) Evaluate the limits of that conclusion—sampling bias,

[1] Code is available at `https://github.com/lohjo/CAEM3A_Screen_Time_Project`

measurement error, and unadjusted confounding under Pearl's back-door criterion—to establish why the finding supports an association rather than a causal effect.

2. Literature Review

This section synthesizes findings from the emerging empirical literature to examine the causal pathways through which ChatGPT use affects total hours dedicated to schoolwork.

2.1. **Time Compression and Efficiency Gains.** A dominant finding across multiple studies is that ChatGPT use consistently reduces the time students need to complete specific academic tasks, primarily by compressing information retrieval, writing, and problem-solving processes. Students frequently report that ChatGPT enables them to accomplish assignments more rapidly than they could using traditional methods such as Google searches, library research, or independent writing.

2.2. **Cognitive Load Reduction.** The theoretical foundation for understanding ChatGPT's effect on time allocation lies in Cognitive Load Theory. ChatGPT appears to reduce both extraneous and intrinsic cognitive load, enabling faster completion of academic tasks. A randomized controlled trial examining ChatGPT-enhanced mobile messaging in online learning (n=63 graduate students over 16 weeks) found that the AI tool reduced the cognitive resources consumed by information management, thereby freeing working memory capacity for germane processing [8]. The treatment group showed significantly higher behavioral engagement (partial $\eta^2 = 0.21$, $p < 0.001$), with 85% of messages being on-task compared to 80% in the control group [8]. This suggests that the time-saving effect operates through a reduction in cognitive overhead associated with searching, organizing, and managing information.

2.3. **Time reallocation toward deeper engagement.** A crucial nuance is that time saved on routine tasks may be reallocated to more cognitively demanding activities rather than resulting in a net reduction in total schoolwork time. The quasi-experimental study by Shaikh et al. found that ChatGPT use under instructor guidance positively affected students' autonomous motivation ($\beta = 0.551$, $p < 0.01$) and engagement ($\beta = 0.506$,$p < 0.01$), which in turn promoted self-directed learning ($\beta = 0.481$, $p < 0.01$) and improved research skills [1]. The direct effect of ChatGPT on research skills was non-significant; rather, the effect was entirely mediated through motivation, engagement, and self-directed learning [1]. This implies that ChatGPT does not simply reduce time spent—it transforms the quality of time invested by fostering greater autonomy and deeper cognitive processing.

2.4. **The risk of reduced independent work time.** However, a countervailing concern is that ChatGPT may reduce the time students spend on independent critical thinking and exploratory research. In the qualitative study of learners' experiences, participants expressed that ChatGPT could lead to "surface-level understanding of topics due to overreliance," "hindrance to the development of independent research skills," and "loss of connection with in-depth learning" [2]. One participant noted, "I worry that solely relying on ChatGPT for information retrieval could compromise the depth of understanding and critical analysis necessary in academic research" [2]. Another student warned, "It may kill the creativity of students, and it will increase the number of students with the attitude of waiting to be spoon-fed" [4]. These concerns are echoed in the Chinese survey, where 47.1% of college students believed that "spending time using ChatGPT may lead to dependency on it," and the tool was seen to potentially "aggravate the phenomenon of knowledge fragmentation," where learners "only know the answer, but not the reasons why" [5].

2.5. **Limitations and Extension.** The current literature has several important limitations. First, most studies rely on self-reported time estimates rather than objective time-tracking data, raising concerns about accuracy. Second, the cross-sectional and quasi-experimental designs preclude strong causal inference; only one study reviewed employed a randomized controlled trial [8]. Third, the studies are predominantly conducted in higher education settings, limiting generalizability to K-12 contexts. Fourth, few studies directly measure total hours spent on schoolwork as a primary outcome variable—most measure efficiency, perceived usefulness, or learning outcomes, and infer time implications secondarily. Future research should employ longitudinal designs with objective time-use metrics (e.g., digital tracking) to establish causal effect sizes more precisely.

Ultimately, ChatGPT use causally reduces the time required for specific academic tasks through cognitive load reduction and information access efficiency. However, the net effect on total hours spent on schoolwork is contingent on key moderating factors: the quality of student engagement (active vs.

passive), the presence of instructor guidance, the nature of the task, and students' self-regulatory capacities. Under optimal conditions, time saved on routine tasks can be reallocated to higher-order thinking, self-directed learning, and deeper engagement. Under suboptimal conditions, ChatGPT may reduce total schoolwork time by enabling shortcut behaviors that bypass meaningful cognitive processing. The relationship must therefore be understood not as a simple univariate causal effect but as a complex, moderated mediation process interwoven with motivational, pedagogical, and dispositional factors.

3. Data collection and survey design

3.1. **What the estimand requires.** For a back-door adjustment set Z to exist and be usable, the instrument must supply four things: a quantitative or ordinal measure of AI use X; a quantitative measure of total study time Y, not time on a single task and not a Likert perception item; at least one pre-treatment measure of competing time demands W; and at least one pre-treatment measure of ability, motivation or course difficulty A. Variables caused by X, such as time saved or perceived academic benefit, must be measurable but must *not* enter Z when the total effect is wanted.

Our own survey supplies X and Y and neither W nor A. That gap is the reason Section 5 exists, and it is a design failure we record rather than repair.

3.2. **Data collection and survey design.** The instrument was a Google Form of nine items, circulated between 11 and 21 July 2026 within the researcher's own course and social networks. Fifteen students responded. Table 1 lists the items and their analytic role.

#	Item	Variable	Role
1	Current year of study	`year`	Background
2	School / course cluster	`school`	Background
3	Hours per week using AI tools (e.g. ChatGPT, Claude, Gemini) for schoolwork	`ai_hours`	**Explanatory** X
4	Hours per week of independent study time (e.g. no AI tools)	`study_hours`	**Response** Y
5	AI tool(s) used most (select all)	`tools`	Construct check
6	Primary use of AI tools (select one)	`use`	Construct check
7	Academic confidence, 1–10	`confidence`	Intended covariate
8	"I would struggle academically without AI tools", 1–5	`dependence`	Intended covariate
9	Submission timestamp (automatic)	`ts`	Administrative

Table 1. Survey instrument. Items 3 and 4 requested a plain number and supplied a worked example ("e.g. 3.5") to discourage range answers.

This is a convenience sample and not a probability sample. The distinction is not a formality. The sampling frame is unknown, no respondent has a calculable inclusion probability, and the guarantee that a sample mean is unbiased for a population mean comes from the sampling design, which is absent. The realised composition shows the recruitment route: six of fifteen respondents are from one school (SOE) and six are in Year 3, against roughly five per year for an evenly spread sample of a three-year diploma. Self-selection compounds this, since an optional survey about AI most attracts students with strong views about AI in either direction. The sign of the resulting non-response bias cannot be determined, because enthusiasts over-reporting use and sceptics over-reporting study time push the association in opposite directions. Nothing in the analysis repairs any of this, so no statement below is generalised beyond the fifteen respondents observed.

3.3. **Primary survey design and collection.** This section describes the participants and demographics, survey design, data collection process, and analysis methods used to examine AI tool use on the number of independent schoolwork hours in Ngee Ann Polytechnic.

3.3.1. *Participants and demographics.* The data come from a convenience sample of n=15 Ngee Ann Polytechnic students recruited via the researcher's own networks, yielding no known sampling frame, no calculable inclusion probabilities, and no design-based warrant for unbiasedness — a fact compounded by visible over-representation of senior SOE students and by self-selection that likely amplifies strong opinions on AI. Both variables are retrospective self-reports of weekly hours, introducing three further issues: large recall error for diffuse activities like studying; opposing social-desirability biases that may understate AI use (perceived as shortcutting) and overstate study time; and definitional ambiguity, since students who intermittently consult a chatbot while studying have no principled way to classify that time. The most extreme observation (37 hours/week) is plausibly a unit error or item misinterpretation, but cannot be verified and is therefore retained in the primary analysis with its influence examined separately; free-text contamination in multiple-choice fields further confirms that the response options failed to capture actual behaviour. Because nothing in the analysis corrects these problems, no claim is extended beyond the fifteen respondents actually observed.

3.4. **Processing the responses.** Four decisions were required before analysis, and two of them changed reported numbers.

- *Blank cells.* Four of the fifteen respondents left exactly one field blank each: one skipped year, one school, one the primary-use item, and one the dependence rating. Both `ai_hours` and `study_hours` are complete for all fifteen, so the test in Section 4 uses the full sample with no deletions and the blanks restrict only subgroup work.
- *An invalid category.* One respondent selected "Year 0", which is not a year of study at this institution. The row is retained because both hour responses are usable, and the respondent's year is treated as unknown rather than imputed.
- *A parsing defect.* One respondent answered "None" to the tools item. Read with default settings, `pandas` coerces the string `None` to a missing value, which converts a substantive answer into a hole and inflates the blank count from four to five. Re-reading with `keep_default_na=False` restores it. "None" is a legitimate answer to a tool-use question and the default configuration destroys it; the earlier draft of this analysis reported one missing value in `tools` for exactly this reason, and the correct count is zero.
- *Free text in closed fields.* Three respondents typed prose into the multiple-choice tools item ("Copilot cuz Its built in", "None, Deepseek sometimes", "None, If really needed then Google AI overview") and one answered the primary-use item with a description of group work under time pressure. These were not recoded into existing categories, because recoding would conceal the finding, which is that the option list did not span the respondents' actual behaviour. They are counted separately in Table 2. The raw export was not modified; all cleaning operates on a copy.

3.5. **All nine items.** Table 2 summarises the categorical items and Table 3 the quantitative and ordinal ones.

Item	Category	n
Year of study	Year 1 / Year 2 / Year 3	2 / 5 / 6
	"Year 0" (invalid) / blank	1 / 1
School cluster	SOE / HS / HMIS	6 / 3 / 2
	BA / ICT / LSCT / blank	1 / 1 / 1 / 1
Tools (24 mentions, 15 respondents)	ChatGPT / Gemini / Claude	8 / 8 / 4
	DeepSeek / Copilot / Google AI Overview	2 / 1 / 1
	answered "none" then named a tool	3
Primary use	Solving problems / Explaining concepts	5 / 4
	Checking or writing work / Translation or summarising	2 / 2
	Free text / blank	1 / 1

TABLE 2. Categorical items. Three respondents answered "none" to the tools item and then named a tool, indicating that "AI tool use" was not understood uniformly; those respondents are likely to under-report `ai_hours`.

Three pairwise relationships are worth recording, none significant at 5% and none of which could plausibly have been at $n = 15$. The relationship this paper is about, taken on the original continuous scale of X, is positive: Pearson $r = 0.358$ ($p = 0.191$; Spearman $\rho = 0.360$, $p = 0.188$). More AI use goes with *more* independent study, not less, which is the direction the median-split test in Section 4 also finds while discarding information the correlation retains.

Respondents reporting more AI hours rated their academic confidence lower, $r = -0.432$ ($p = 0.108$). Cross-sectional data cannot separate the two readings of this: low confidence plausibly drives students

Variable	n	Mean	Median	SD	Min	Max
`ai_hours` (h/week)	15	3.02	2.00	2.61	0	8
`study_hours` (h/week)	15	10.33	6.00	9.63	2	37
`confidence` (1–10)	15	5.40	6.00	2.35	1	8
`dependence` (1–5)	14	2.29	2.50	1.14	1	4

TABLE 3. Quantitative and ordinal items. Two respondents reported exactly 0 AI hours and a third 0.25, so the sample contains genuine non-users. The standard deviation of `study_hours` is almost as large as its mean, which is the main reason the test in Section 4 is inconclusive.

to lean on AI, and heavy reliance on AI plausibly erodes confidence. Both fit equally well, and the ambiguity is the subject of Section 4.2.

The `dependence` item, which was collected in order to serve as a covariate, tracks confidence weakly (Spearman $\rho = 0.239$, $p = 0.411$) and AI hours not at all ($\rho = -0.033$, $p = 0.911$). Whatever it measures, it is not a usable proxy for prior ability, and Section 4.2 gives a second and stronger reason not to adjust for it.

3.6. **Measurement.** Both X and Y are self-reported retrospective estimates of weekly hours, which carries three problems. Recall error is expected to be large, because respondents reconstruct diffuse activity such as "studying" poorly. Social-desirability pressure plausibly acts in opposite directions on the two items, understating AI use, which may feel like an admission of shortcutting, and overstating study time; the net effect on the observed association cannot be signed. Definitional ambiguity is the sharpest of the three: item 4 asks for study time using "no AI tools," but a student who consults a chatbot intermittently during a three-hour session has no principled way to classify those hours, and the three respondents in Table 2 who answered "none" and then named a tool demonstrate that the construct was not read uniformly.

The largest observation, 37 hours per week of independent study, is a plausible unit error (a day read as a week) or a misreading of the item rather than a genuine value. With the survey closed it cannot be verified, so it is retained in the primary analysis and its influence is quantified in Section **??** instead of being assumed away.

3.7. **The sampling model.**

Assumption 1. Conditionally on the observed vector of AI hours, and hence on the induced partition into groups of sizes n_1 and n_2, the response values satisfy

$$Y_{1,1}, \dots, Y_{1,n_1} \overset{\text{iid}}{\sim} \mathcal{N}(\mu_1, \sigma^2), \qquad Y_{2,1}, \dots, Y_{2,n_2} \overset{\text{iid}}{\sim} \mathcal{N}(\mu_2, \sigma^2),$$

with the two samples mutually independent, $\sigma^2 \in (0, \infty)$ unknown and common to both groups, and $\mu_1, \mu_2 \in \mathbb{R}$ unknown.

Assumption 1 bundles four requirements: independence within each group, independence between groups, normality of each group, and homoscedasticity. Independence between groups holds by construction, since each respondent contributes to exactly one group. Independence within groups is a design claim that convenience sampling makes debatable, since respondents recruited through overlapping social and course networks may share courses, deadlines and study habits, and it is the one assumption for which no diagnostic is available. The remaining two are testable and both are examined in Section **??**, where neither survives comfortably.

3.8. **Hypotheses.** Write μ_1 for the mean weekly independent study time in the population of low AI-use students and μ_2 for the corresponding mean among high AI-use students. The hypotheses are

$$H_0 : \ \mu_1 = \mu_2 \qquad \text{against} \qquad H_1 : \ \mu_1 \neq \mu_2,$$

tested two-tailed at $\alpha = 0.05$. No directional prediction was registered before the data were inspected, and substitution and complementarity predict opposite signs, so selecting a one-tailed alternative after seeing which way the sample means fell would inflate the true type I error rate above its nominal level.

A pooled two-sample t-test is used. The population variances are unknown and must be estimated, which rules out a z-test; both groups are small ($n_1 = 8$, $n_2 = 7$), so the central limit theorem does not deliver approximate normality of the standardised difference; and the two groups are independent by construction.

4. Hypothesis test and results

Write μ_1 for the mean weekly independent study time in the population of low AI-use students and μ_2 for the corresponding mean among high AI-use students. The hypotheses are

$$H_0 : \ \mu_1 = \mu_2 \qquad \text{against} \qquad H_1 : \ \mu_1 \neq \mu_2,$$

tested two-tailed at level $\alpha = 0.05$.

4.1. Computation and decision.

The observed data are given in Table 4.

Group	n	Observed `study_hours` (h/week)	Mean	SD
Low AI use (≤ 2 h)	8	$2, 3, 5, 5, 6, 7, 8, 20$	7.0000	5.6061
High AI use (> 2 h)	7	$3, 5, 5, 10, 19, 20, 37$	14.1429	12.1714

Table 4. Group summaries. Sample variances are $s_1^2 = 31.4286$ and $s_2^2 = 148.1429$.

The observed difference in means is $\bar{y}_1 - \bar{y}_2 = 7.0000 - 14.1429 = -7.1429$ hours per week. The pooled variance is

$$s_p^2 = \frac{7(31.4286) + 6(148.1429)}{13} = \frac{220.0000 + 888.8571}{13} = \frac{1108.8571}{13} = 85.2967, \qquad s_p = 9.2356,$$

the standard error is

$$\text{SE} = s_p\sqrt{\tfrac{1}{8} + \tfrac{1}{7}} = 9.2356\sqrt{0.267857} = 9.2356 \times 0.517549 = 4.7799,$$

and therefore

$$t = \frac{-7.1429}{4.7799} = -1.4944, \qquad \nu = 8 + 7 - 2 = 13.$$

The two-tailed critical value is $t_{13,0.975} = 2.1604$, so H_0 is rejected if $t < -2.1604$ or $t > 2.1604$. The observed value lies inside the acceptance region, and the corresponding p-value is

$$p = 2\,\mathbb{P}(T_{13} > 1.4944) = 0.1590 > 0.05.$$

At the 5% level there is insufficient evidence to conclude that mean weekly independent study time differs between low and high AI-use students.

4.2. Confounding.

The limitation that matters most is structural rather than distributional, and no amount of robustness checking touches it. The two groups compared differ in far more than their AI use, and the test conditions on nothing at all.

Pearl's back-door criterion states the requirement precisely [10]. In a causal graph G, a set Z satisfies the back-door criterion relative to an ordered pair (X, Y) if

(i) no node in Z is a descendant of X, and
(ii) Z blocks every path between X and Y that begins with an arrow into X.

When such a Z exists and is observed, the interventional distribution is identified from observational data by the adjustment formula

$$\mathbb{P}(y \mid do(x)) = \sum_z \mathbb{P}(y \mid x, z)\,\mathbb{P}(z), \tag{1}$$

and a comparison of groups within levels of Z estimates a causal effect.

The two-sample t-test is exactly the case $Z = \varnothing$, and $\varnothing$ satisfies the criterion only if there is no back-door path at all, that is, no common cause of AI use and study time. That is implausible here, and this survey's own data show why. Workload and course demands plausibly raise both AI use and total study hours, opening $X \leftarrow \text{workload} \rightarrow Y$; no workload item was collected. Prior ability and motivation plausibly do the same, and the `confidence` item, a weak proxy for them, correlates with `ai_hours` at $r = -0.432$, which is the signature of an open path. Year of study and course cluster plausibly affect both, and the sample is markedly unbalanced on both. Every one of these paths is left open, so the difference in Table 4 confounds the effect of AI use with the effects of whatever else distinguishes heavy from light users.

Adjustment is not available here for two reasons. The methods used are bivariate and cannot accommodate covariates, so no multivariable model can be fitted. More fundamentally, the two additional variables that were collected are unfit for the purpose. The `dependence` item, agreement with "I would struggle academically without AI tools," is plausibly a *consequence* of AI use rather than a pre-treatment cause of it, which makes it a descendant of X and a mediator on the causal path. Conditioning on it would violate condition (i) directly, block part of the very effect being estimated, and can additionally induce collider bias if the mediator shares an unobserved cause with Y; this is the "bad control" pattern catalogued by Cinelli *et al.* [2], and Section 5.4 demonstrates it numerically. Adjusting for the available covariate would therefore make the estimate worse, not better.

Measurement error compounds the problem in a way adjustment could not fix even if covariates were available: classical non-differential error in a dichotomised X attenuates the observed contrast toward zero, so part of the non-significance may be measurement artefact rather than absence of association.

The consequence for wording is strict. What has been established is that *in this sample, no statistically significant difference in mean independent study time was observed between low and high AI-use students.* It has not been established that AI use does not affect study time, nor that the observed difference estimates any causal quantity.

5. Secondary analysis: back-door adjustment

5.1. **Motivation and dataset selection.** Section 4.2 leaves two claims untested: that a valid adjustment set would change the answer, and that adjusting for a mediator would damage it. Neither can be checked on fifteen respondents with no confounders measured, so the same causal question is put to a public dataset that has the variables this instrument lacks.

Four candidates were assessed against the requirements of Section **??**. The Middlebury survey [3] has the best variable match, including weekly study hours and GenAI frequency for 634 students, but respondent-level microdata is not distributed. The FLoRA dataset [6] supplies fine-grained interaction logs, but its temporal fields are event timestamps within a two-week assignment and measure time on task, not a weekly total. The global ChatGPT survey [9] has 22,963 respondents across 120 countries, but a full-text search of its questionnaire returns no quantitative study-time item, only Likert perception items; sample size cannot rescue a variable the instrument never asked for.

The Colombian study-habits survey [8] survives: 357 undergraduates at the Manizales campus of a Colombian public university across 14 programmes, with complete cases on every required variable, 357 of 357.

5.2. **Graph and adjustment set.** The assumed graph G has node set $\{X, Y, W, A, L, M_1, M_2\}$ and edge set

$$W \to X, \quad W \to Y, \quad A \to X, \quad A \to Y, \quad L \to X, \quad X \to M_1, \quad M_1 \to Y, \quad X \to M_2, \quad M_2 \to Y.$$

Here W is competing time demands, A collects prior ability, motivation and course difficulty, L is AI literacy and access, M_1 is perceived help understanding topics, and M_2 is AI-induced procrastination.

Since W, A, M_1 and M_2 each have degree two with neighbours exactly $\{X, Y\}$, and L has degree one, every simple X–Y path has length two and no path routes through L. The enumeration in Table 5 is therefore complete by construction rather than by sampling.

Path	Back-door?	Blocked by $Z = \{W, A\}$?	Blocked by $Z' = \{W, A, M_1\}$?
$X \leftarrow W \to Y$	yes	yes, fork at $W \in Z$	yes
$X \leftarrow A \to Y$	yes	yes, fork at $A \in Z$	yes
$X \to M_1 \to Y$	no, causal	no, left open	yes, and that is the fault
$X \to M_2 \to Y$	no, causal	no, left open	no

Table 5. Complete enumeration of the four X–Y paths.

$Z = \{W, A\}$ satisfies the criterion. Neither element is a descendant of X, since $\mathrm{de}(X) = \{M_1, M_2, Y\}$; both back-door paths are blocked at their fork; no element is a collider on any X–Y path, so conditioning opens nothing; and both causal paths remain open, so the coefficient on X carries the *total* effect with the efficiency and procrastination channels summed.

$Z' = \{W, A, M_1\}$ fails, and fails on condition (i) rather than condition (ii). The edge $X \to M_1$ places $M_1 \in \mathrm{de}(X)$. Note that Z' still blocks every back-door path, which is exactly why the failure must be diagnosed by (i): M_1 is a mediator, not a confounder.

L lies on no X–Y path, so Z blocks all back-door paths whether or not L is adjoined, and L is omitted. Adjoining it would not be invalid, but L is instrument-like and including instruments can amplify bias from unmeasured confounding [2]. The omission rests on an exclusion restriction: L influences Y only through X. If the true graph contains $L \to Y$, or a latent U with $L \leftarrow U \to Y$, then $X \leftarrow L \to Y$ is an open back-door path and $Z = \{W, A\}$ is insufficient.

The set A is operationalised as three sub-nodes rather than one composite: cumulative GPA, procrastination frequency, and academic programme entered as 13 dummies. This refinement leaves the criterion intact, because each sub-node retains degree two with neighbours $\{X, Y\}$, so the single fork through A is replaced by three parallel forks, each blocked at its middle node. The invariance is conditional and not free: omitting even one sub-node leaves that fork unblocked.

5.3. **Estimation and results.** Y is recorded as six ordered, interval-censored bands with an open top category, so the primary estimator is proportional-odds ordinal logistic regression, with ordinary least squares on band midpoints and HC3 standard errors reported alongside as the only reading interpretable in hours. The right-hand side is identical for both: X, W, and the three sub-nodes of A. This regression adjustment realises (1) under a linear-index restriction, and M_1 and M_2 are excluded so that both causal paths stay open.

Estimator	$\hat{\beta}_X$	SE	95% CI	statistic	p
Ordinal logit (log-odds per step)	0.2158	0.1319	$(-0.0428, 0.4744)$	$z = 1.635$	0.102
OLS on midpoints, HC3 (h/week)	0.1589	0.0944	$(-0.0260, 0.3439)$	$t = 1.684$	0.092
OLS, unadjusted (h/week)	0.0883	0.0965		$t = 0.915$	0.360

Table 6. Back-door-adjusted estimates under $Z = \{W, A\}$, $n = 357$, with the unadjusted estimate for reference. Odds ratio for the ordinal fit is $e^{0.2158} = 1.241$.

Neither adjusted estimate rejects $H_0 : \beta_X = 0$ at $\alpha = 0.05$. Both point estimates are positive: adjusting for competing time demands and ability, more frequent AI use is associated with *more* independent study time, about $+0.16$ hours per week per step on the five-point frequency scale, which is the opposite sign to the substitution hypothesis.

Adjustment for Z nearly doubles the point estimate relative to the unadjusted comparison, from 0.088 to 0.159 hours. The unadjusted figure was not merely imprecise; it was biased toward zero by the confounders. This is the concrete evidence that the confounding described in Section 4.2 is material and not hypothetical, and it is what the t-test of Section 4 is structurally unable to see.

5.4. **Falsification: adjusting for the mediator.** Refitting with $Z' = \{W, A, M_1\}$, on the same data with the same n and one variable added, gives Table 7.

Estimator	$\hat{\beta}_X$ under Z	$\hat{\beta}_X$ under Z'	Change	p under Z'
Ordinal logit	0.2158	0.1017	-52.9%	0.495
OLS, HC3 (h/week)	0.1589	0.0967	-39.2%	0.369

Table 7. Effect of conditioning on the descendant M_1.

Mediator-inclusion bias is demonstrated rather than asserted. Conditioning on M_1 collapses the estimate toward zero in the direction the graph predicts, because path $X \to M_1 \to Y$ of Table 5 is blocked and only the M_2 channel remains in the coefficient. The p-value moves from 0.10 to 0.50. An analyst who "controlled for everything available" would report a substantively different and non-identified result. This is the empirical form of the argument in Section 4.2 for not adjusting on `dependence`.

5.5. **Limitations of the secondary analysis.** The outcome's top band is open-ended and holds 51.26% of the sample, so the midpoint recode must assume a ceiling. Coding it at 5.5, 6.5 and 7.5 hours gives $\hat{\beta}_X = 0.1589$ ($p = 0.092$), 0.1906 ($p = 0.109$) and 0.2222 ($p = 0.127$). The sign and the non-significance are stable and the magnitude scales with the assumed ceiling, which is why the ordinal model, which makes no such assumption, is primary.

W is a partial proxy. The instrument's only competing-demands item distinguishes full-time students from those who study and work, and captures no assignment-load information, so residual workload confounding is not adjusted away and the requirement that Z be measured without error fails.

The procrastination sub-node of A may be post-treatment. The item asks about procrastination in general and does not separate the trait, a pre-treatment cause, from AI-induced procrastination, which is M_2 and a descendant of X. To the extent it measures the latter, including it violates condition (i), the same error demonstrated in Section 5.4.

The graph is asserted from domain reasoning and is not verified from the data; the sample is a single Colombian campus; and the design is cross-sectional and self-reported. What Table 6 reports is an adjusted association that equals a causal effect only under the assumption of no unmeasured confounding.

6. Conclusion

The pooled two-sample t-test gives $t = -1.4944$ on 13 degrees of freedom against a critical value of ± 2.1604, with $p = 0.1590$, so $H_0 : \mu_1 = \mu_2$ is not rejected at $\alpha = 0.05$. Theorem **??** shows the critical value is exact under the stated model, and the median split does not compromise it. The decision survives every relaxation attempted: Welch's test, an exact permutation test free of parametric assumptions, the exact Mann–Whitney test, and deletion of the most influential observation all agree.

The result should be recorded as inconclusive rather than null, for reasons quantified above rather than merely listed. The confidence interval $(-17.47,\ 3.18)$ hours per week excludes no effect of practical interest. Power against the observed effect is 0.283, so failure to detect was the likely outcome regardless of the truth. The point estimate moves by half its value when one respondent is removed. The sample is a self-selected convenience sample of fifteen with no calculable inclusion probabilities, so the inferential machinery has no design basis. Both variables are self-reported recall estimates with an ambiguous response definition and at least one probable unit error, and three respondents demonstrably did not read "AI tool use" as the instrument intended. And the test conditions on the empty set, which does not satisfy the back-door criterion in any plausible graph for these variables.

That last point is the one the secondary analysis makes concrete. On 357 Colombian undergraduates, adjusting for a set that provably satisfies the criterion moves the estimate from $+0.088$ to $+0.159$ hours per week, while adjusting instead for a post-treatment mediator moves it to $+0.097$ and the p-value from 0.10 to 0.50. The difference between a valid and an invalid adjustment set is larger than the effect being estimated. A bivariate test that conditions on nothing cannot report a causal quantity, however many robustness checks it passes.

Both samples point the same way, away from substitution: high AI users in the primary sample studied 7.1 hours per week more on average, and the adjusted secondary estimate is $+0.159$ hours per week per step of AI-use frequency. Neither reaches significance, they differ in country, scale, measurement and adjustment, and the primary estimate is far too unstable to quantify. Two samples agreeing is mildly suggestive and nothing more.

Five changes would make a future iteration informative. Raise the sample to at least 28 per group, as the power calculation requires for the observed effect size. Replace convenience recruitment with probability sampling stratified by school and year, which would also correct the SOE and Year-3 over-representation. Measure time objectively where possible, through device screen-time reports or learning-management-system activity logs, rather than by recall. Collect pre-treatment confounders explicitly, namely assignment count, contact hours and prior grade point average, so that a set satisfying the back-door criterion can be formed and a multivariable model fitted, and analyse X on its original continuous scale rather than dichotomised, which recovers the power lost in Remark **??**. Finally, pilot the questionnaire to repair the option lists that respondents overrode with free text, define "independent study" so that intermittent AI use has a clear home, and add range validation to reject implausible hour values at entry.

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